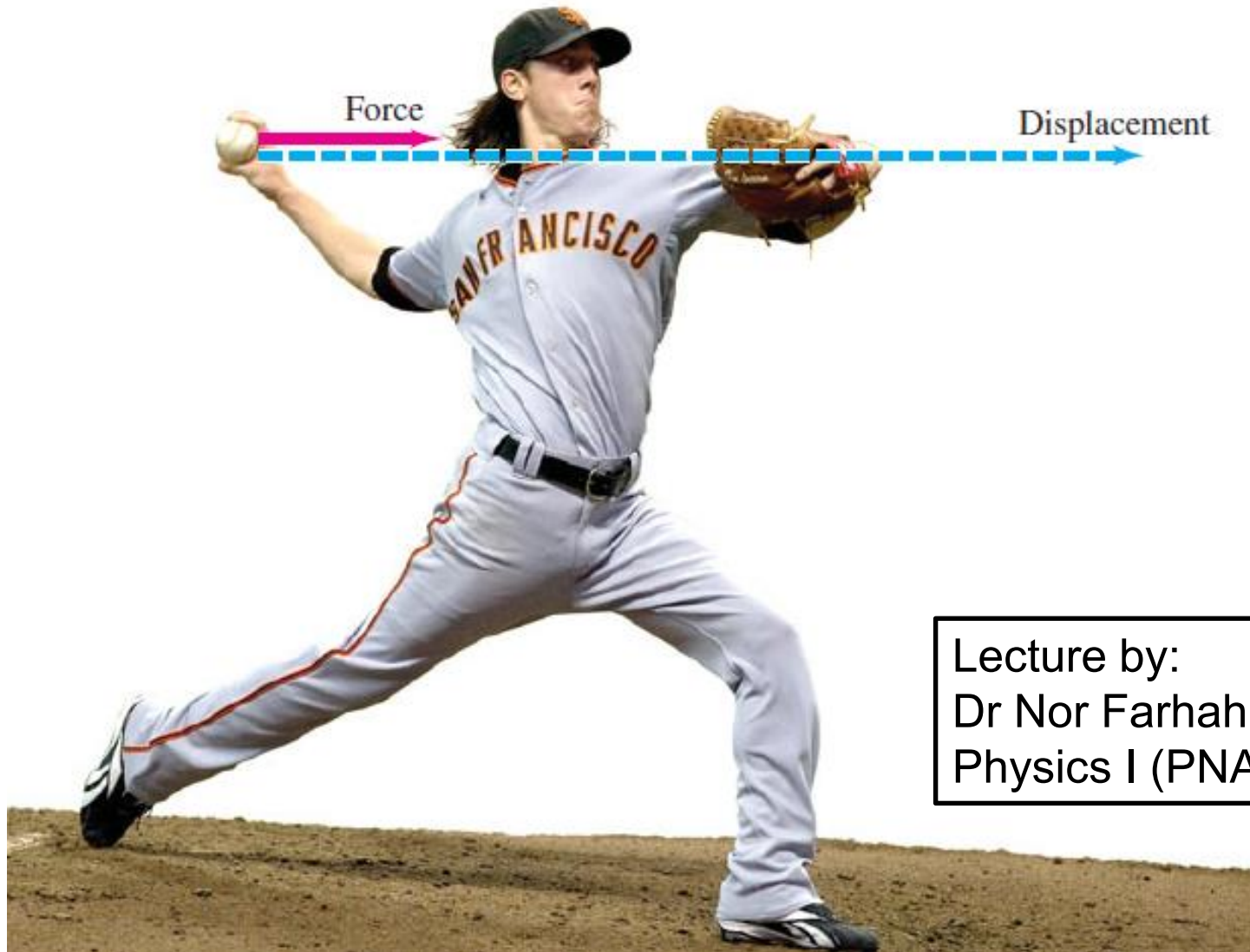


Chapter 6: Work and Energy



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Physics I (PNAP0123)

Chapter 6: Work and Energy

Sub-Topics:

- **Work**
- **Kinetic Energy and Energy Theorem**
- **Conservative Forces & Non-conservative Forces**
- **Gravitational Potential Energy**
- **Gravity and Non-conservative Forces**
- **Spring Potential Energy**
- **System & Energy Conservation**
- **Power**
- **Work Done by a Varying Force**

6-1 Work Done by a Constant Force

Determining work done by a force

- When force acts on object, the object move through distance. Defining as work.
- In everyday use, work is usually thought of as effort expended by a body, you, on homework, or on a job.
- In physics, we define work W as force F times the displacement s , over which the force acts:
- Force is parallel to the direction of the displacement.

$$W = Fs$$

work done by a constant force

- Unit Newton-meter (N m) a Joule (J).

Determining work done by a force

$$W = Fs$$

work done by a constant force

EXAMPLE: Find the work done by the 25-Newton force F in displacing the box $s = 15$ meters.



SOLUTION:

FYI •The units of (N m) are Joules (J). You can just keep them as (N m) if you prefer.

6-1 Work Done by a Constant Force

Determining work done by a force

$$W = Fs$$

work done by a constant force

- If the force is not parallel to the displacement the formula for work has the minor correction

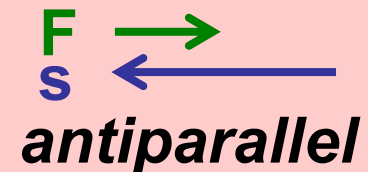
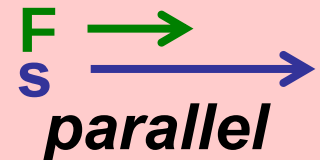
$$W = Fs \cos \theta$$

work done by a constant force
not parallel to displacement

Where θ is the angle between **F** and **s**.

FYI

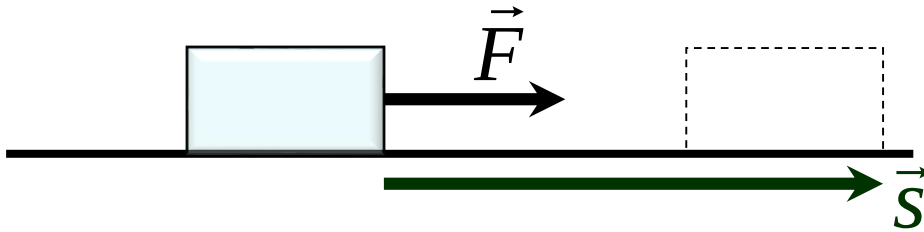
- If F and s are parallel, $\theta = 0^\circ$
and $\cos 0^\circ = +1$.
- If F and s are antiparallel, $\theta = 180^\circ$
and $\cos 180^\circ = -1$.



Applications of work's equation

Case 1 :

► Work done by a horizontal force, F on an object.

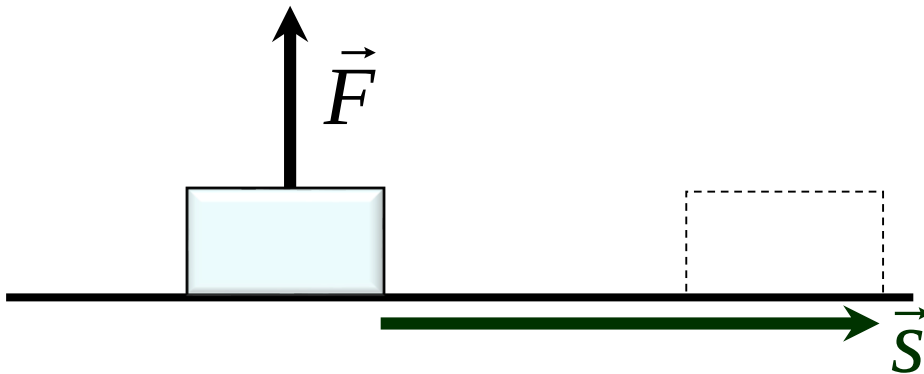


$$W = Fs \cos \theta \quad \text{and} \quad \theta = 0^\circ$$

$$W = Fs$$

Case 2 :

Work done by a vertical force, F on an object.



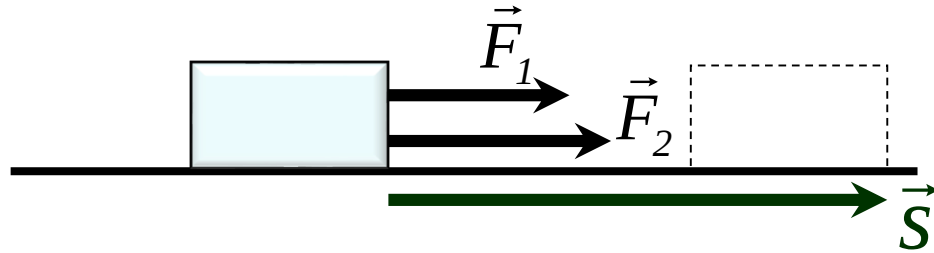
$$W = Fs \cos \theta \quad \text{and} \quad \theta = 90^\circ$$

$$W = 0 \text{ J}$$

Applications of work's equation

Case 3 :

Work done by a horizontal forces, F_1 and F_2 on an object



$$W_1 = F_1 s \cos 0^\circ$$

$$W_2 = F_2 s \cos 0^\circ$$

$$\sum W = W_1 + W_2 = (F_1 s + F_2 s)$$

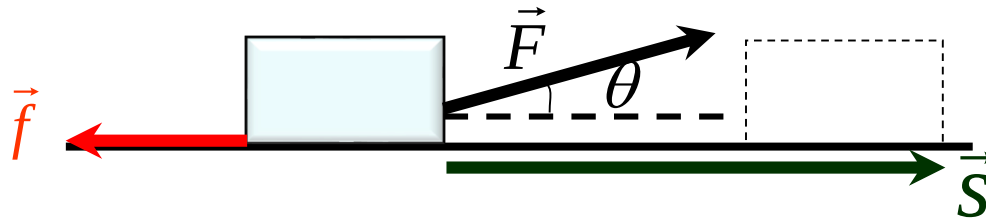
$$\sum W = (F_1 + F_2) s \quad \text{and} \quad F_{\text{nett}} = F_1 + F_2$$

$$\sum W = W_{\text{nett}} = (F_{\text{nett}}) s$$

Applications of work's equation

Case 4 :

Work done by a force, F and frictional force, f on an object



$$W_{nett} = (F_{nett})s \quad \text{and} \quad F_{nett} = F \cos \theta - f = ma$$

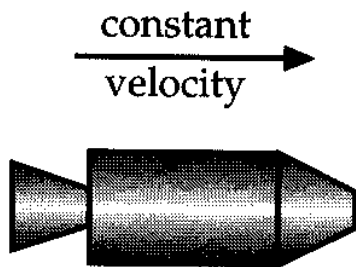
$$W_{nett} = (F \cos \theta - f)s \quad \text{OR} \quad W_{nett} = mas$$

► Caution :

Work done on an object is zero when $F = 0$ (a) or $s = 0$ (b) and $\theta = 90^\circ$ (c).

$$F = 0$$

work done = 0
on spacecraft

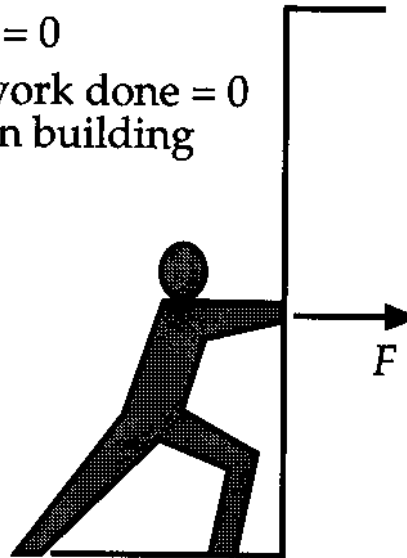


Spacecraft
travelling with
constant velocity

(a)

$$s = 0$$

work done = 0
on building

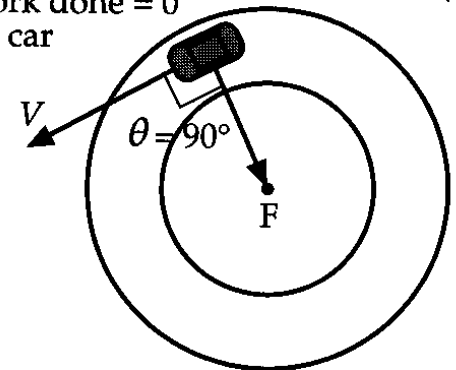


Force exerted on
a building

(b)

$$\theta = 90^\circ$$

work done = 0
on car



Centripetal force acting
on the car moving
with constant speed in
a circle

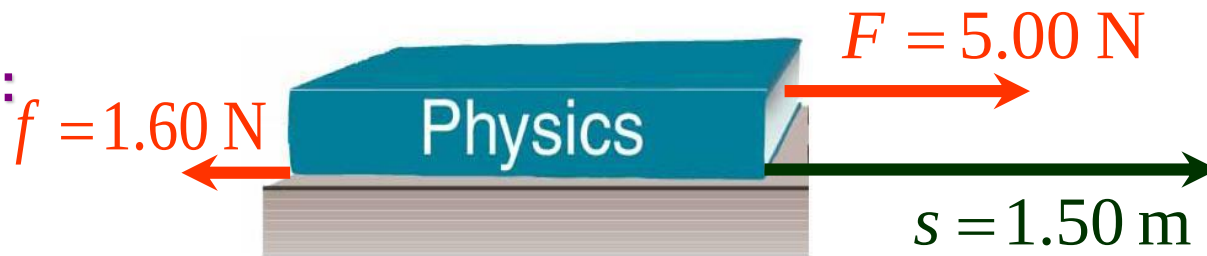
(c)

EXAMPLE:

You push your reference book 1.50 m along a horizontal table with a horizontal force of 5.00 N. The frictional force is 1.60 N. Calculate

- the work done by the 5.00 N force,
- the work done by the frictional force,
- the total work done on the book.

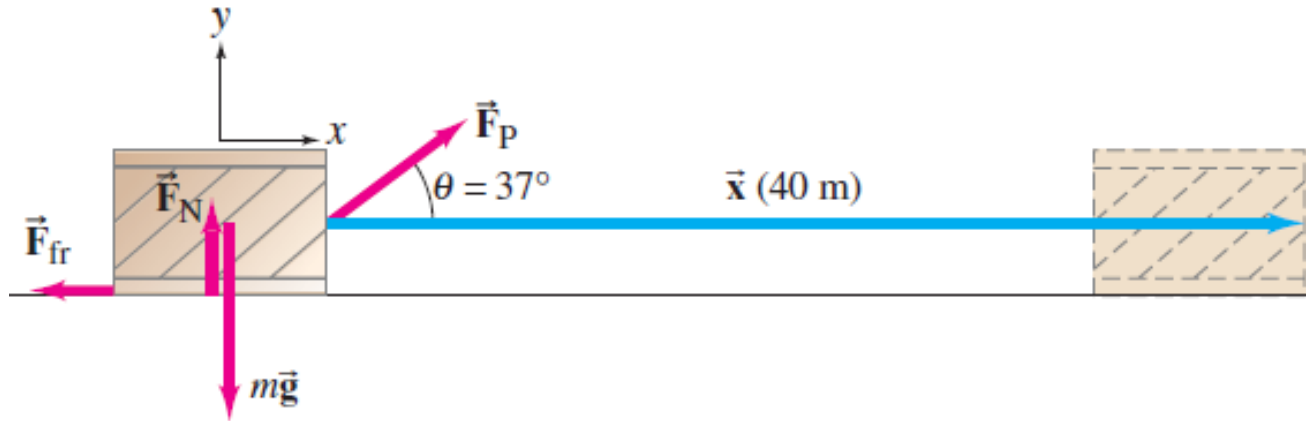
SOLUTION:



- Use work's equation of constant force,

Solving problems

A person pulls a 50-kg crate 40 m along a horizontal floor by a constant force $F_p = 100 \text{ N}$, which acts at a 37° . The floor is rough and exerts a friction force $F_{fr} = 50 \text{ N}$. Determine (a) the work done by each force acting on the crate, and (b) the network done on the crate.



EXERCISE:

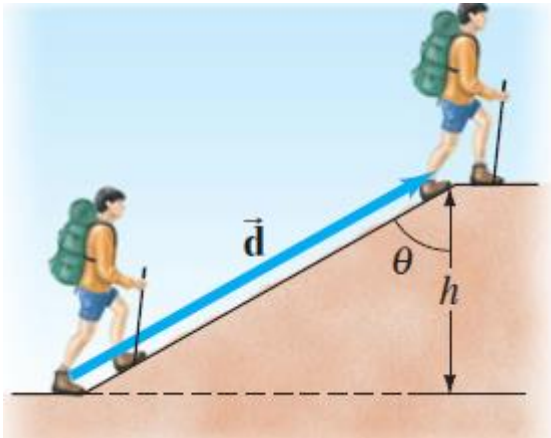
A block of mass 2.50 kg is pushed 2.20 m along a frictionless horizontal table by a constant 16.0 N force directed 25.0° below the horizontal. Determine the work done on the block by

- a. the applied force,
- b. the normal force exerted by the table, and
- c. the gravitational force.
- d. Determine the total work on the block.

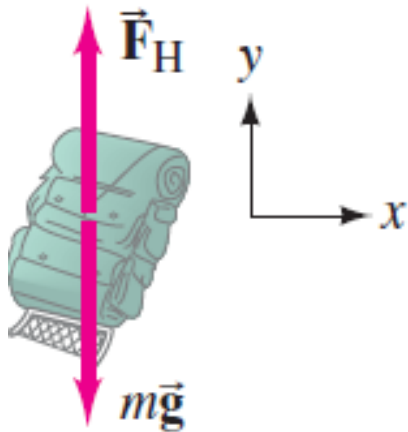
ANS. : 31.9 J; (b) & (c) U think; 31.9 J

SOLVING PROBLEM

- (a) Determine the work a hiker must do on a 15.0-kg backpack to carry it up a hill of height $h=10.0$ m, as shown in Fig.
- (b) Determine also the work done by gravity on the backpack, and
- (c) the net work done on the backpack. For simplicity, assume the motion is smooth and at constant velocity (i.e., acceleration is zero).

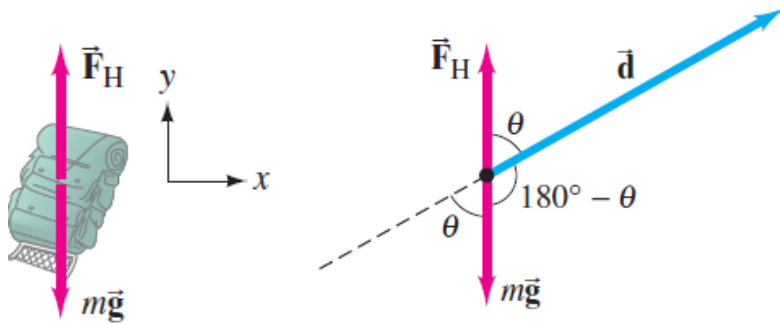


1. Draw a free-body diagram
2. Choose a coordinate system
3. Apply Newton's laws.

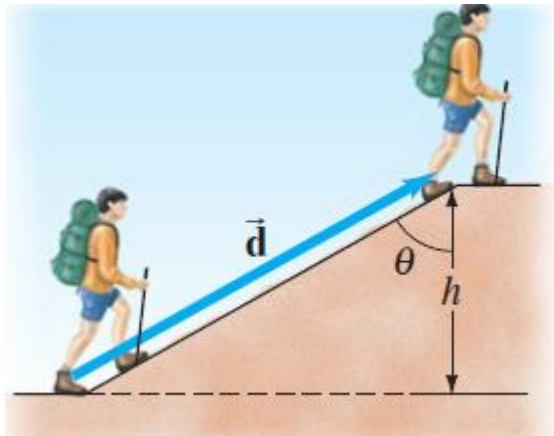
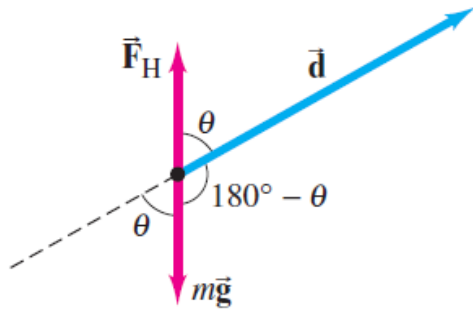


SOLVING PROBLEM

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- (c) the net work done on the backpack. For simplicity, assume the motion is smooth and at constant velocity (i.e., acceleration is zero).



(b) The work done by gravity on the backpack is



(c) The *net* work done on the backpack is

6-2 Kinetic Energy and Work-Energy Theorem

Kinetic energy

- **Kinetic energy** KE is the energy of motion.
- The bigger the speed v , the bigger KE .
- The bigger the mass m , the bigger KE .
- The formula for KE , justified later, is

$$KE = (1/2)mv^2$$

kinetic energy

- Looking at the units for KE we have

$$\text{kg}(\text{m/s})^2 = \text{kg m}^2 \text{s}^{-2} = (\text{kg m s}^{-2})\text{m}.$$

- In the parentheses we have a mass times an acceleration which is a **Newton**.
- Thus, KE is measured in (N m), which are (J).
- An object in motion has the ability to do work and thus can be said to have energy

6-2 Kinetic Energy and Work-Energy Theorem

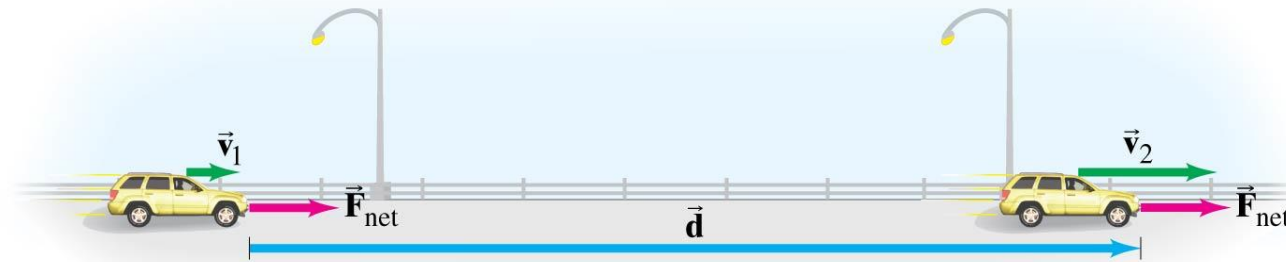


FIGURE 6-7 A constant net force F_{net} accelerates a car from speed v_1 to speed v_2 over a displacement d . The net work done is $W_{\text{net}} = F_{\text{net}} d$.

$$a = \frac{v_2^2 - v_1^2}{2d} \quad \text{Eq. 2-11c } (v_2^2 = v_1^2 + 2ad)$$

$$W_{\text{net}} = F_{\text{net}} d, \quad F_{\text{net}} = ma$$

$$W_{\text{net}} = F_{\text{net}} d = mad = m \left(\frac{v_2^2 - v_1^2}{2d} \right) d = m \left(\frac{v_2^2 - v_1^2}{2} \right)$$

$$W_{\text{net}} = \frac{1}{2} m v_2^2 - \frac{1}{2} m v_1^2$$

$$\text{KE} = \frac{1}{2} m v^2.$$

We define this as translational KE

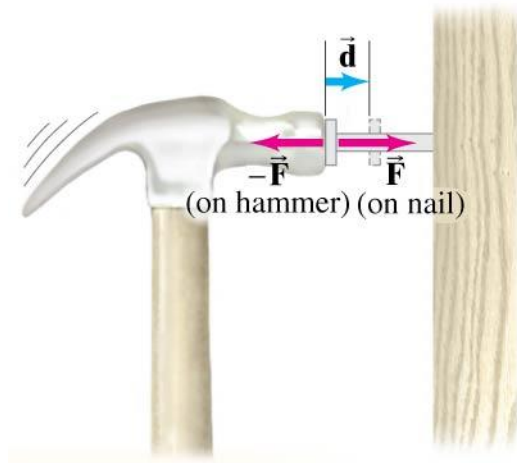
6-2 Kinetic Energy and Work-Energy Theorem

- This means that the work done on an object is equal to the change in object's kinetic energy: (Work-energy principle !!!)

$$W_{\text{net}} = \Delta_{\text{KE}} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2.$$

- If the net work is **positive**, the kinetic energy **increases**.
- If the net work is **negative**, the kinetic energy **decreases**.

6-2 Kinetic Energy and Work Energy Theorem



A moving hammer strikes a nail and comes to rest. The hammer exerts a force F on the nail; the nail exerts a force $-F$ on the hammer (Newton's third law).

The work done on the nail by the hammer is positive ($W_n = Fd > 0$). The work done on the hammer by the nail is negative ($W_h = -Fd$).

The decrease in kinetic energy of the hammer is equal to the work the hammer can do on the nail .

****Because work and kinetic energy can be equated, they must have the same units: kinetic energy is measured in joules.**

6-2 Kinetic Energy and Work Energy Theorem

Determining work done by a force

EXAMPLE: Find the work done by the brakes in bringing a 730-kg Smart Car to a rest in 80. meters if its starting speed is 32 m/s.



SOLUTION:

6-2 Kinetic Energy and Work Energy Theorem

Work done as energy transfer

$$W_{net} = \Delta KE$$

Work energy principle

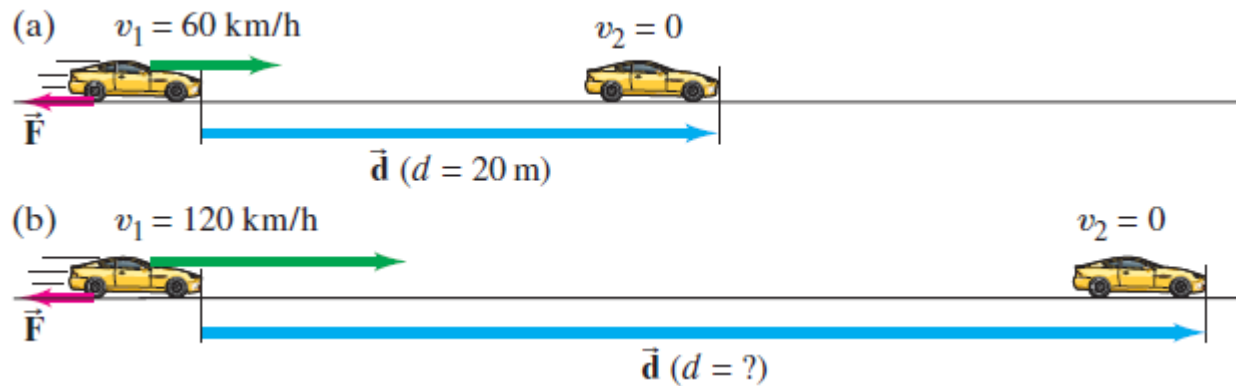
EXAMPLE: Use energy to find the work done by the brakes in bringing a 730-kg Smart Car to a rest in 80. meters if its starting speed is 32 m/s.



SOLUTION:

Solving problems:

A car traveling 60 km/h can brake to a stop in a distance $d=20$ (Fig. 6–10a). If the car is going twice as fast, 120 km/h, what is its stopping distance (Fig. 6–10b)? Assume the maximum braking force is approximately independent of speed.



6-3 Conservative and Non-conservative Forces*

There are two types of forces:

1. Conservative Forces

A force is called conservative if the work it does does not depend on the path taken, but only on the initial and final positions of the object.

Eg. *Gravitational force, Spring force, Electrostatic force*

2. Non-Conservative Forces

A force is called non-conservative if the work it does depends on the path taken.

E.g. *Friction force, Air resistance (drag), Applied push/pull forces with energy loss*

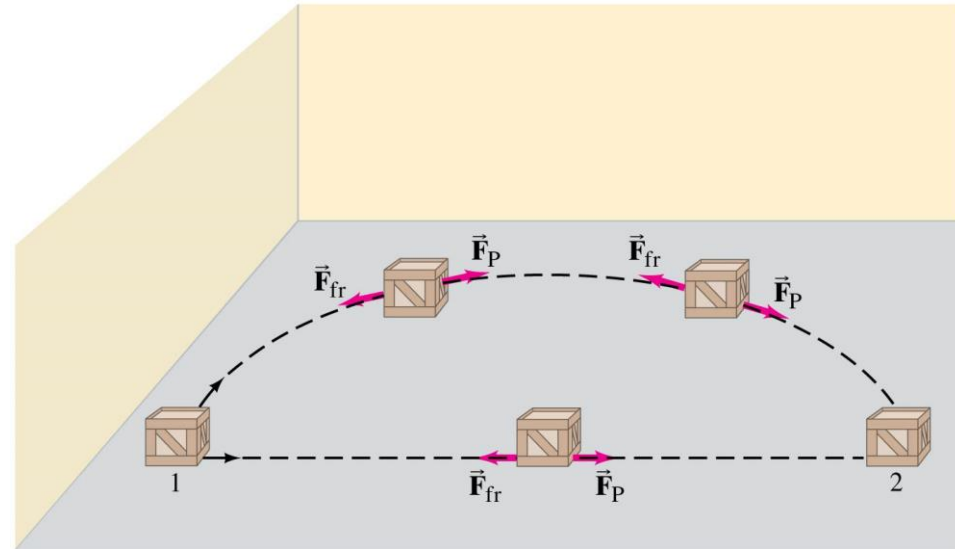
6-3 Conservative and Nonconservative Forces

TABLE 6-1 Conservative and Nonconservative Forces

Conservative Forces	Nonconservative Forces
Gravitational	Friction
Elastic	Air resistance
Electric	Tension in cord
	Motor or rocket propulsion
	Push or pull by a person

Potential energy can only be defined for conservative forces.

If friction is present, the work done depends not only on the starting and ending points, but also on the path taken. Friction is called a nonconservative force.



6-3 Conservative and Nonconservative Forces

Therefore, we distinguish between the work done by conservative forces and the work done by nonconservative forces.

$$W_{\text{net}} = W_{\text{C}} + W_{\text{NC}}.$$

Then, from the work-energy principle, Eq. 6–4, we have

$$W_{\text{net}} = \Delta \text{KE}$$

$$W_{\text{C}} + W_{\text{NC}} = \Delta \text{KE}$$

where $\Delta \text{KE} = \text{KE}_2 - \text{KE}_1$. Then

$$W_{\text{NC}} = \Delta \text{KE} - W_{\text{C}}.$$

Work done by a conservative force can be written in terms of potential energy, as we saw in Eq. 6–7b for gravitational potential energy:

$$W_{\text{C}} = -\Delta \text{PE}.$$

We combine these last two equations:

$$W_{\text{NC}} = \Delta \text{KE} + \Delta \text{PE}. \quad (6-10)$$

Thus, *the work W_{NC} done by the nonconservative forces acting on an object is equal to the total change in kinetic and potential energies.*

6-3 Conservative and Nonconservative Forces

If there are no nonconservative forces, the sum of the changes in the kinetic energy and in the potential energy is zero—the kinetic and potential energy changes are equal but opposite in sign.

$$W_{\text{NC}} = \Delta \text{KE} + \Delta \text{PE}$$

$$\Delta \text{KE} + \Delta \text{PE} = 0$$

conservative
forces only

$$(\text{KE}_2 - \text{KE}_1) + (\text{PE}_2 - \text{PE}_1) = 0.$$

This allows us to define the total mechanical energy:

$$E = \text{KE} + \text{PE}$$

$$\text{KE}_2 + \text{PE}_2 = \text{KE}_1 + \text{PE}_1$$

And its conservation:

$$E_2 = E_1 = \text{constant.}$$

6-4 Potential Energy

- ▶ is defined as the energy stored in a body or system because of its position, shape and state.

Gravitational potential energy, U

- ▶ is defined as the energy stored in a body or system because of its position.

- ▶ Equation :

$$U = mgh$$

where

U : gravitational potential energy

m : mass of a body

g : acceleration due to gravity

h : height of a body from the initial position

- ▶ The gravitational potential energy depends only on the height of the object above the surface of the Earth.

6-4 Potential Energy

- Work-gravitation potential energy

◦ For calculation, use

$$W = \Delta U = U_f - U_i$$



where

U_f : final gravitational potential energy

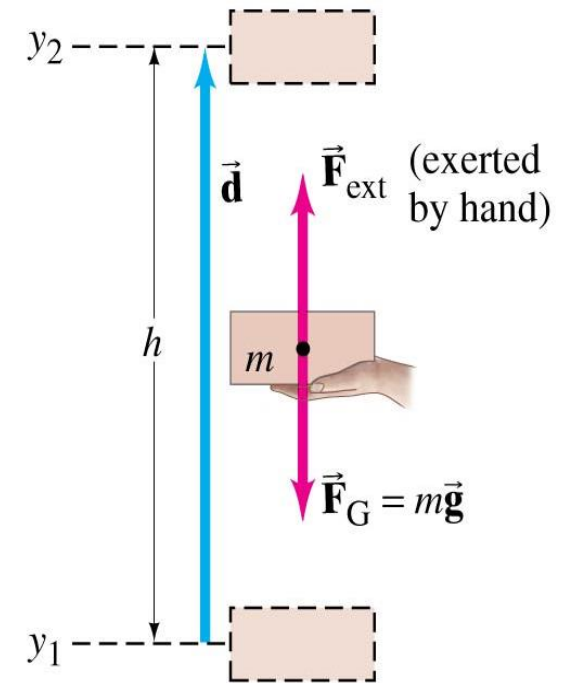
U_i : initial gravitational potential energy

W : work done by a gravitational force

6-4 Potential Energy (PE or U)

1) In raising a mass m to a height h , the work done by the external force is

$$\begin{aligned} W_{\text{ext}} &= F_{\text{ext}} d \cos 0^\circ = mgh \\ &= mg(y_2 - y_1) \end{aligned}$$



Work done by gravity:

$$W_G = F_G d \cos \theta = mgh \cos 180^\circ.$$

$$\begin{aligned} W_G &= -mgh \\ &= -mg(y_2 - y_1). \end{aligned}$$

Work done lifting the object has been stored as GPE

$$\text{PE}_G = mgy.$$

GPE due to Earth's gravity = object's weight mg x height y

Allow the object to fall freely under the action of gravity. it acquires

$$\text{velo } v^2 = 2gh$$

After falling a height, it has KE $\rightarrow \frac{1}{2} mv^2 = mgh$

$$\text{PE}_G = mgy.$$

The higher an object is above the ground, the more GPE it has.

$$W_{\text{ext}} = mg(y_2 - y_1)$$

$$W_{\text{ext}} = \text{PE}_2 - \text{PE}_1 = \Delta \text{PE}_G.$$

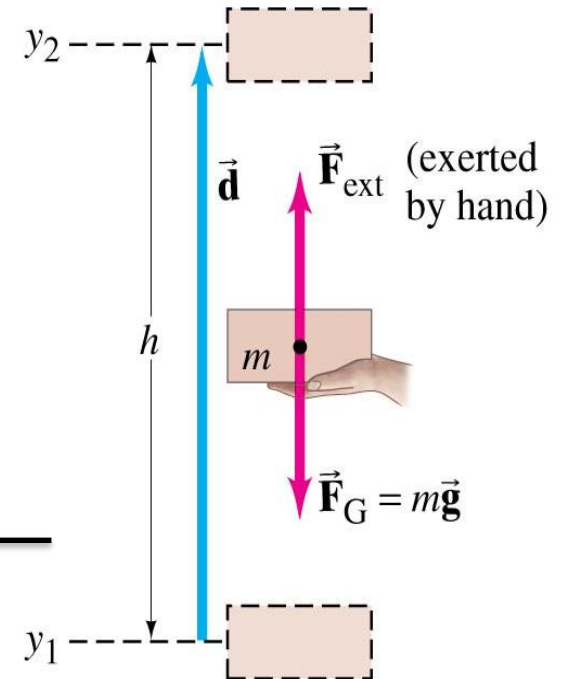
Δ in PE_G when an object moves from a y_1 to $y_2 = W_{\text{ext}}$ move the object from position 1 to 2 without acceleration.

$\Delta \text{PE}_G =$ negative of the work done by the gravity itself.

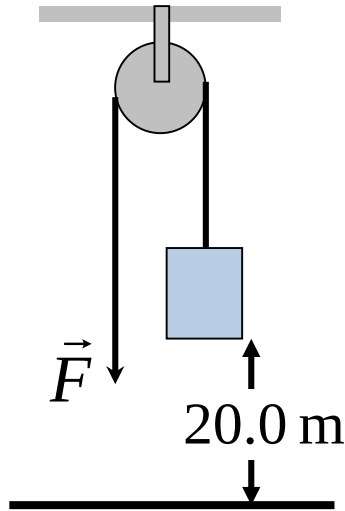
$$W_G = -mg(y_2 - y_1)$$

$$W_G = -(\text{PE}_2 - \text{PE}_1) = -\Delta \text{PE}_G$$

$$\Delta \text{PE}_G = -W_G.$$



Example :



In a smooth pulley system, a force F is required to bring an object of mass 5.00 kg to the height of 20.0 m at a constant speed of 3.00 m s^{-1} as shown in Figure.

Determine

- the force, F
- the work done by the force, F .

Solving problem

A 1000-kg roller-coaster car moves from point 1, to point 2, and then to point 3. (a) What is the gravitational potential energy at points 2 and 3 relative to point 1? That is, take $y=0$ at point 1. (b) What is the change in potential energy when the car goes from point 2 to point 3? (c) Repeat parts (a) and (b), but take the reference point ($y=0$) to be at point 3.

